

The Meaning of Choreographies: Making a Case for a Multimodal Semantic Analysis

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Abstract

Empirical observations have shown that there are clear intuitions of dance-music coherence in dance performances. This paper builds on previous applications of formal linguistic methodology to other cognitive systems to assess the advantages of considering a semantic analysis over a purely syntactic approach, to account for dance and music perceptual integration as a coherent whole. Syntactic analyses of music and of dance have been proposed, and were recently combined into a case-study analysis identifying structural matches/mismatches between dance and music. Formal semantic analyses of narrative dance, of music, and of combined music-animations suggest that semantics may also play a role in dance-music inferences. Despite support from both formal and non-formal research, the joint meaning of combined dance and music has not been formally addressed. Based on choreographical examples, I first examine how dance-music multimodal interpretations can be analyzed purely syntactically by mapping congruent (corresponding) structural properties. However, comparison with intuitive interpretations of dance-music sequences demonstrates that a strictly syntactic analysis is too narrow to account for the complex range of inferences evoked by choreographies. I then assess how previous applications of formal semantic methodology to non-standard objects – namely truth-conditional semantics and discourse referents – can extend to dance-music combinations, and further propose a novel analysis of multimodal sequences using semantic conjunction of each modality’s set of denotations. I argue that such a semantic analysis is not only possible, but also necessary to better account for intuitive inferences by distinguishing cases of dance-music in-/coherence within both syntactic incongruence and congruence. Finally, I conclude that a combined semantic analysis of dance-music sequences is necessary and complementary to the syntactic congruence analysis in order to productively account for intuitive inferences of coherence. As a more general goal, this work attempts to offer a novel, formal approach to current endeavors exploring the underlying cognitive architecture of integrative perceptual processing by investigating multimodal interactions.

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1 Introduction

The present paper examines perceptual integration of combined dance and music from syntactic and semantic approaches. Watching a dance performance is a rich perceptual experience that induces complex mental representations, and one doesn’t need to be a professional musician or dancer to feel the strong connection between music and movement. Research across a large range of disciplines from neuroscience to dance theory highlights an intricate relationship between sound and movement extending to music and dance¹. Though appealing to distinct modalities and perceptual systems, it has been empirically observed that there are clear intuitions about the coherence of dance performances as a merged combination² (Fogelsanger and Afanador, 2006; Howlin et al., 2017; Krumhansl and Schenck, 1997; Mitchell and Gallaher, 2001; Vincs et al., 2010). Despite a rapidly developing body of research, mechanisms of dance-music integration are poorly understood.

Dance and music are claimed to be means of human communication that can connect through structure and/or meaning. Formal syntactic descriptions of structure perception in music (Lerdahl and Jackendoff, 1983) and in dance (Charnavel, 2019), as well as dance with music have highlighted instances of dance-music syntactic relations at a rhythmic level (Charnavel, 2023). Despite a lack of consensus as to whether dance and music communicate semantic content, there is convincing work in formal semantics on referentiality of music and of narrative dance (Patel-Grosz, Katz, et al., 2022; Migotti Ramponi, 2023; Schlenker, 2022). Still, there have been few formal attempts inspired by linguistics to address syntactic and semantic representations of joint dance and music.

Building from previous applications of formal linguistic methodology to other cognitive systems, the main goal of this paper is to assess the advantages of a semantic analysis over a purely syntactic approach to formally account for intuitions of dance-music coherence³. More globally, studying dance and music integration may shed light on the general underlying principles of multimodal perceptual processing of audio-visual stimuli, and on structural representations and meaning creation in human means of communication (Patel-Grosz, Grosz, et al., 2022).

Question

Specifically, the question we ask is this: if indeed there are clear intuitions about the coherence of combined dance and music, are they based on syntactic processes only, or do semantic processes also contribute to it? This yields two hypotheses:

- **Hypothesis 1 (Syntax Only):** If our intuitive sense of dance-music coherence solely arises from syntactic mapping, we’d expect syntactic congruence to lead to perception of dance-music coherence, and syntactic incongruence to incoherence. Alternatively, if our sense of coherence is not solely elicited by structural congruence, we would observe cases in which syntactic congruence is perceived as incoherent and syntactic incongruence as coherent.
- **Hypothesis 2 (A role for semantics):** If Hypothesis 1 doesn’t hold, perhaps the semantic combination of dance and music plays a role in our intuitive sense of coherence. If so, we’d expect semantic comprehensibility of joint dance and music to better predict coherence than syntax, thus explaining discrepancies between intuitions and syntactic predictions. Alternatively, if semantics is not involved in dance-music integration, it would inaccurately predict coherence.

¹Considering the diversity of dance performances and their interplay with aesthetics, “music” and “dance” will be defined simplistically as be subsets of all possible sounds and of all possible movements, respectively.

²For the present purposes, “combined dance-music” will be used as a simplified functional definition of a “choreography”, synonymously to “dance performance”.

³In/coherence judgments do not make assumptions about aesthetics, liking, pleasantness of a choreography, which are subject to individual experience and cultural background (e.g. Monroy et al., 2022).

Claim

The main claim is that our intuitive perception of coherence cannot be solely explained by a structural (syntactic) analysis, but that it must be complemented by a combined semantic analysis of dance and music, which is not only *possible* but also *necessary* to understand dance-music perception, integration, and interpretation.

To support this claim, I will be analyzing extracts from real-life dance performances. In some works, dance and music are structurally very congruent and appear coherent – such as Balanchine’s choreographies. Other works exhibit drastically different structural features between dance and music, and feel incoherent – such as the first part of Naharin’s “Venezuela” (1).

(1) Naharin’s “Venezuela”⁴

The same dance is repeated twice with a different music. The dance is composed of very rhythmic dynamic moments and some more fluid sequences. The music in Part 1 is quite unstructured (flowing gregorian chants with few contrasts and an unclear tempo). The music in Part 2 is a rhythmic, intense electronic style of music with a clear beat and tempo. Although Part 2 is structurally congruent on average, Part 1 is clearly incongruent across many structural parameters – and feels incoherent.

A first approximation is therefore that syntactic in/congruence between dance and music seems to predict in/coherence, but the picture is not so clear-cut. Simply think about the structural incongruence of ongoing music with stillness in dance or of silence with dance movements; they are pervasive in choreographies but aren’t necessarily associated with incoherence. Taking the “Angry-Happy” fictional example (Fig.1⁵), although it is undoubtedly structurally congruent, the anger acted out and the chanting joy just feel intuitively incoherent. Thus, syntactic congruence doesn’t imply coherence. A semantic evaluation, however, would yield incomprehensibility and thus adequately predict the feeling of incoherence.



Figure 1: “Angry-Happy” fictional example: A single dancer with clenched fists stomps heavily on the stage – as if they were angry – to the beat of a joyful, grandiose, light-hearted music (Haendel’s “Halleluia” for example, fn.5). The dancer changes direction at each musical phrase boundary.

Intuition: Incoherent.

Syntax: Congruent (the dancer and music share the same beat and phrase boundaries).

Semantics: Incomprehensible, even contradictory.

By contrast, consider Cunningham’s choreography “Beach Birds” (Fig.2). Despite the lack of structural matches between the dance and music, it feels intuitively coherent. Thus, syntactic incongruence doesn’t imply incoherence. Semantics plays a role here: if dance can refer to *birds* and music to a

⁴A trailer for Naharin’s “Venezuela”, performed by the Batsheva Dance Company, can be found here: <https://youtu.be/HsHjQbNeEIU?feature=shared>. The background music of this trailer is that of Part 1.:

⁵An example of joyful music that could accompany this dance sequence is Haendel’s “Halleluia” (https://www.youtube.com/watch?v=PHcJ_95WgNw).

seashore, then together they can evoke *birds on a seashore* (Fig.2b), which is semantically comprehensible.



(a) Screenshot taken from a video extract of Cunningham’s “Beach Birds”, 1991, which can be found at https://www.youtube.com/watch?v=0IH_rrpj0CU.



(b) A pictorial depiction of the situation *birds on a seashore*. (Picture in public domain.)

Figure 2: **Merce Cunningham’s “Beach Birds”**: Dancers make contrasted awkward movements meant to mimic the stiff, clumsy, and fluttering motion of seabirds on the ground, with temporally fluid movements (Fig.2a). The accompanying music by John Cage involves rain-sticks and is temporally fluid as well, but doesn’t share any structural parallels with the dance.

Intuition: Coherent.

Syntax: Incongruent.

Semantics: Comprehensible; it could represent the situation depicted in Fig.2b, for example.

Outline

To put this work into its context, Section 2 briefly reviews the literature of “informal” experimental investigations and “formal” theories of dance and music using linguistic methodology. Section 3 explores Hypothesis 1 – that dance-music coherence arises strictly from a syntactic analysis of joint dance-music sequences based on congruent mapping of structural properties, and how it may be too narrow to account for our perception of choreographies. I then argue in Section 4 that an additional semantic layer (Hypothesis 2) is possible and necessary to offer a more fine-grained account of the complex range of inferential effects that arise from choreographies by building on tools from formal semantics. Finally, this paper concludes that relying both on syntax *and* on semantics is necessary to understand the sense of coherence between dance and music. Although there are limitations to this approach, it provides tools for more fine-grained typologies of dance-music inferences and may be a productive line of research in formal linguistics, cognition, and the arts.

2 Investigations of Dance and Music: State of the Art

Some claim that our perception of dance and music is composed of solely structural and modality-internal information, while others hold that dance and music have meaning, in that they have the ability to make reference to an external reality. The nature of inferences evoked by dance and music have been addressed both “informally” (Section 2.1), and “formally” with theoretical approaches from the field of linguistics applied to non-standard objects (Section 2.2).

2.1 Informal Accounts

We start from the observation that there is a “robust sense of congruency and incongruency” (Mitchell and Gallaher, 2001) between music and dance that it is consistent across viewers (Fogelsanger and Afanador, 2006; Howlin et al., 2017; Krumhansl and Schenck, 1997; Vincs et al., 2010). Dance-music coherence seems to be dependent on the significant relationships between dance and music, but it is unclear at what level (structural and/or semantic) this sense of coherence takes place.

So, what produces this “matching”, “congruency”, or “incongruency”, “mismatch”, “disunity”, “dissonance”? In the fields of experimental cognitive psychology and neuroscience, a number of studies have delineated some reliable structural (such as tempo, dynamics, contour, large-scale organization of units), emotional (intensity, tension), imagery (such as texture, roundness, choppiness), spatial (such as coming closer), energy level, and motor parallels between music and movement (e.g. Eitan and Granot, 2006; Godøy et al., 2016; Kelkar and Jensenius, 2018; Zatorre et al., 2007; Zhou et al., 2015), and music and dance more specifically (e.g. Christensen et al., 2016; Clarke, 2005; Howlin et al., 2017; Johnson and Larson, 2003; Karpati et al., 2015; Karpati et al., 2017; Krumhansl and Schenck, 1997). Theoretical work on choreomusical mappings (e.g. Hodgins, 1992) also contributes to the general agreement for the existence of a strong dance-music relationship with a diversity of comparable features.

Intriguingly, even when dance and music do not “map” to each other, this incongruence doesn’t necessarily lead to incoherence (Fogelsanger and Afanador, 2006; Mitchell and Gallaher, 2001), and there is something special about their very combination (Howlin et al., 2017)⁶. Coherence may therefore be achieved without explicit correspondences between dance and music. It appears that compatibility and merging of meaning may also be at play, such that dance and music can emphasize, complement, and compete with each other (Cook, 1998).

Dance-music relationships therefore take two forms: mappings from one modality to another and emergent combinations. Whereas the former is dependent on defining what “similarity” consists of (Godøy et al., 2016), the latter is fundamentally anchored in overall perceived unity, and specifically semantic unity (comprehensibility). This relies on the ability of dance and music inferences to influence each other into some sort of emergent holistic experience that is more than the sum of its parts. Dance and music are not simply juxtaposed in our perception the same way they are in the physical reality (Fogelsanger and Afanador, 2006).

Yet, the diversity of definitions and measurements claimed to be attributed to “semantics” and “syntax” in dance and music (e.g. Howlin et al., 2017; Krumhansl and Schenck, 1997; Mitchell and Gallaher, 2001; Vincs et al., 2010) reveal a lack of theoretical frameworks guiding how to differentially test for structural or semantic strategies. Consequently, many accounts fall short of the fine-grained features that determine multimodal coherence. I hope that a formal analysis inspired by linguistic methodology can help decipher some of these conceptual confusions by offering tools to tackle issues of syntax and meaning in multimodal sequences.

2.2 Formal Accounts

Let us now briefly overview of foundational theories of syntax and semantics of dance, of music, and of mixed sequences that the present paper builds on.

⁶Howlin et al. (2017) hold that the meaning-making process is particularly prominent in cases of incongruence, that could still be perceived as “matching” in some way. In their study, they found arousal responses to the dance-music condition that wasn’t in each modality separately. However, they only measured physiological arousal and continuous ratings of enjoyment but not intuitive coherence *per se*.

Syntax

Pioneered by Lerdahl and Jackendoff (1983), syntactic approaches to dance and to music have aimed to build a basic grammar of these media from an analytical description⁷. In their “Generative Theory of Tonal Music” (GTTM), Lerdahl and Jackendoff stipulate that a musical grammar can be divided in two main dimensions: rhythmical structure (grouping and meter) and tonal pitch space. Hypothesizing that syntactic structures of dance and music share abstract properties, Charnavel (2019) proposed to transpose certain modules of GTTM into a theory of dance syntax and specifically rhythmic components (namely grouping and meter). Grouping rules are mainly grounded in general cognition and Gestalt principles, such as principles of proximity, similarity, and closure (Wertheimer, 1938), guiding how we assemble auditory events together into chunks. Amongst other well-formedness rules, grouping configurations must be contiguous and non-overlapping. Preference rules outline which structural interpretations of a sequence would be more likely to be perceived over other well-formed structures.

In her case study, Charnavel (2023) compared the syntactic analyses of the music and dance sequences of three choreographies (by Nijinsky, Béjart, and Bausch) danced on the same musical extract (*Augurs of Spring*, in Stravinsky’s “Rite of Spring”⁸) in terms of their respective grouping and metrical structures. She identified “matches” and “mismatches” between their structures, and showed that structurally ambiguous sequence may be disambiguated by the other modality, preferring matching syntactic interpretations. This hints that rhythm could hypothetically be involved in creating a sense of coherence in dance-music sequences. From these accounts, it remains unclear whether music/dance communicate meaning beyond modality-internal relations, as in whether they are referential – can provide information about a music-/dance-external reality.

Semantics

The hypothesis of a semantics of combined dance and music takes from recent exploratory research demonstrating referentiality of pictures (Abusch, 2012), of music (especially paired to images; Schlenker, 2022), and of narrative dance (Patel-Grosz, Grosz, et al., 2022), with increasing support from experimental evidence (Migotti Ramponi, 2023). However, referentiality of combined dance and music has seldom been addressed.

What we consider as “meaning” here is that of a “rule-governed way in which music can provide information (i.e. license inferences) about some music-external [and dance-external] reality, no matter how abstract” (Schlenker, 2022), that is, how we distinguish between situations of which a sequence is true and situations of which this sequence isn’t true⁹. A recurring issue raised concerning non-standard objects is that of underspecification and abstractness, accompanied by much ambiguity (discussed in detail in Schlenker, 2022). Yet, both informal and formal research demonstrate patterns suggestive of a rule-based system compared to randomly generated semantics. In other terms, even if there are infinitely many possible inferences, some inferences are more likely to arise than others. As a simple example of semantics in music, a decrescendo could denote an object moving further away or going towards a lower energy state (Schlenker, 2022). As for dance, a dancer sitting down may denote an object undergoing a sitting action (Patel-Grosz, Grosz, et al., 2022) or going towards a lower energy state.

When combining different modalities, Schlenker (2022) shows how formal semantics applied beyond language is a fruitful method to explore meaning of music with film/cartoons and hypothesizes that

⁷This description is not meant to be prescriptive about aesthetics, but rather to shed light on the “unconscious knowledge that allows an observer to organize and make coherent the surface patterns of dance” and of music (Charnavel, 2023).

⁸Videos of these extracts are linked in Charnavel’s (2023) paper.

⁹As per previous linguistic approaches, judgments of intuitive “acceptability” (i.e. comprehensibility, plausibility) of possible inferences will be grounded in introspection.

these mixed semantic principles can be extended to combined dance and music. It already appears in Schlenker’s work that the semantics of music can deal with cases that syntax cannot account for. Typically, some musical phrases appear to be naturally linked (such as re-expositions of a theme) even if they are non-contiguous and thus cannot form syntactic constituents, as defined by grouping principles. From a semantic viewpoint, coindexation of discourse referents (i.e. pronouns in language) can more appropriately account for these distal relationships between audio-visual events.

Moreover, supporting the relevance of using formal analyses to investigate multimodal coherence, it has been claimed that speech with gesture also shows patterns of perceptual coherence, that can be attributed to the appraisal of an “intelligible whole” from multimodal semantic relationships pertaining to discourse referents (Lascarides and Stone, 2009). Although the dance-gesture and music-speech analogies are controversial, their work can be paralleled with the observation that there are strong intuitions of dance-music coherence, substantiating our hypothesis that dance-music may also be evaluated holistically using meaning-based multimodal coherence.

Putting non-formal and formal literature together, it clearly stands out that there is a strong cooperation between music and dance. It is however unclear whether their coherence (also referred to as “congruence” or “match”) arises strictly from structural mappings, or if dance-music meaning interactions are also at play. Although there are reasons to hypothesize the requirement for a formal semantic analysis of multimodal phenomena to complement syntactic approaches, not only have choreographies not yet been defined from a formal semantic perspective, but the very possibility of such a semantics has not yet been formally established, hence the aim of this paper. To anchor our analysis, we hereafter assume theories of syntax and semantics of dance and music independently.

3 Theory 1: Syntactic Analysis of Dance-Music Sequences

Considering the general consensus towards a syntax of combined dance and music, we’ll first examine the hypothesis of a purely syntactic analysis of dance-music coherence without assuming any semantic contribution.

Defining Dance-Music Sequences

In the present analysis, “sequences” will constitute our basic unit¹⁰. A sequence is an extract of dance and/or music, composed of individual “streams” (e.g. single monophonic music sequence, single group of dancers with synchronized movements)¹¹, simultaneously superimposed (e.g. several groups of dancers making distinct movements, several concurrent musical voices). We define a dance-music sequence S as being composed of co-occurring sequences of dance and of music: $S = \langle M, D \rangle$.

3.1 Syntactic Mapping

We’ll first explore some cases of dance-music syntactic congruence – the mapping of structural parameters of one modality to another, then turn to cases of syntactic incongruence – non-correspondence between structural parameters, in relation to their predictions of in/coherence.

¹⁰The issue of the actual length of a sequence/event is beyond the scope of this paper.

¹¹Stream segregation in music involves auditory scene analysis (Bregman, 1994; McAdams et al., 2022). Streaming in dance is grounded in Gestalt principles and visual scene analysis, as in Group III of the extract presented in Charnavel’s (2023) analysis of Bausch’s *Augurs of Spring* in which there are two dance streams with a group doing the same movements and a spatially distinct solo dancer doing different movements (Fig.3a).

3.1.1 Congruence and Perceptual Effects

Some structural features are abstract enough to bridge physical differences between modalities (Char-navel, 2023; Eitan and Granot, 2006). Correspondences between these syntactic features between dance and music can induce coherence, such as the following cases:

Syntactic mapping of onset, and duration. Synchronized temporal characteristics such as onset and duration between sound and movement induce structural congruence (Godøy et al., 2016).

Syntactic mapping of music dynamics with visual density. Dynamics in music sometimes map to the amount, intensity or velocity of movements, or to the number of dancers on stage. For example, in Bausch’s “Rite of Spring”, sparser musical events and softer dynamics often parallel relative stillness in the dance (as in 15:49¹²). This echoes observations of quicker movements accompanying crescendos in Balanchine’s “Divertimento N15”¹³ (Krumhansl and Schenck, 1997) or in Béjart’s “Boléro”¹⁴:

Syntactic mapping of accents. It is very common for musical accents (sharp contrast in a musical parameter) to be synchronized with a drastic change of acceleration in movement. For example, in Bausch’s “Rite of Spring”, a jump of the male dancers concurs with two musical accents, synchronized respectively with the impulse off the ground and the landing (13:19, fn.12), enhancing the dance-music relation. In Balanchine’s “Divertimento N15”, dancers follow placement of accents in the score with more emphasized movements (Krumhansl and Schenck, 1997).

Syntactic mapping of rhythm (grouping, tempo, and meter). In her article, Charnavel (2023) specifically identifies the presence of common grouping syntactic features and (to a lesser extent) metrical features between a musical extract from Stravinsky’s “Rite of Spring”, and that of three corresponding choreographies. For example, clear section boundaries in the music are regularly marked by clear section boundaries in the dance in Bausch’s choreography, such as in the example she gives of a solo dancer separating herself from a group, co-occurring with a change in instrumentation (Fig.3). Balanchine’s “Divertimento N15” also contains high congruence of section boundaries and hierarchical organization of units (Krumhansl and Schenck, 1997; e.g. Fig.4). Moreover, musical tempo very often matches that of dance. Among many examples, one can cite the dancers’ small bounces congruent with the tempo (5:56, fn.12) in Bausch’s “Rite of Spring”, or dancers skipping in Part 2 of Naharin’s “Venezuela” (1), perfectly synchronized with the dynamic musical beat.

Syntactic mapping of form: repetitions. When recurring structural patterns (i.e. repetitions of themes or movements) map across modalities, it induces a sense of coherence. For example, in Béjart’s “Boléro”, the same melody-movement pattern periodically re-occurs at each repetition of the theme (Fig.5). In Balanchine’s “Divertimento N15”, there are many repetitions and parallelisms in the music that are congruent with similar repeated movements (e.g. 3:02 and then 3:11¹⁵).

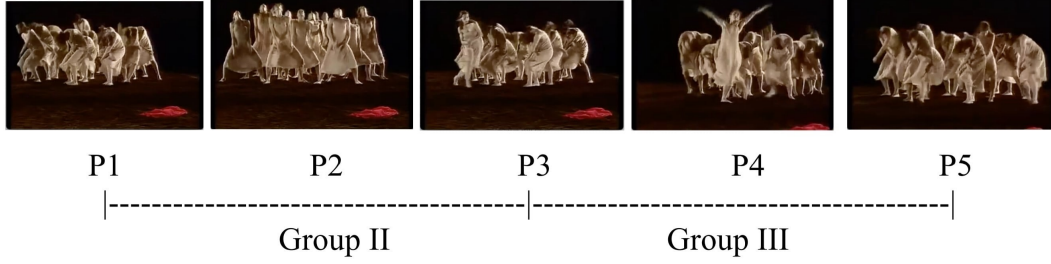
To summarize, the structural parameters that have been claimed to map between dance and music are principally temporal properties including accents, dynamics, rhythm (meter and grouping), and parallelisms – which are syntactic in nature. Syntactic congruence may give the feeling of movement enacting sound and perhaps – more loosely – sound enacting movement. Dance can act as a visual anchor, giving an impression of stability, and can even disambiguate between multiple possible structural interpretations of the music (Charnavel, 2023). So far, syntactic congruence appears to be a good

¹² Timestamps for Bausch’s “Rite of Spring” are taken from <https://www.youtube.com/watch?v=yiWH6vRT4So>.

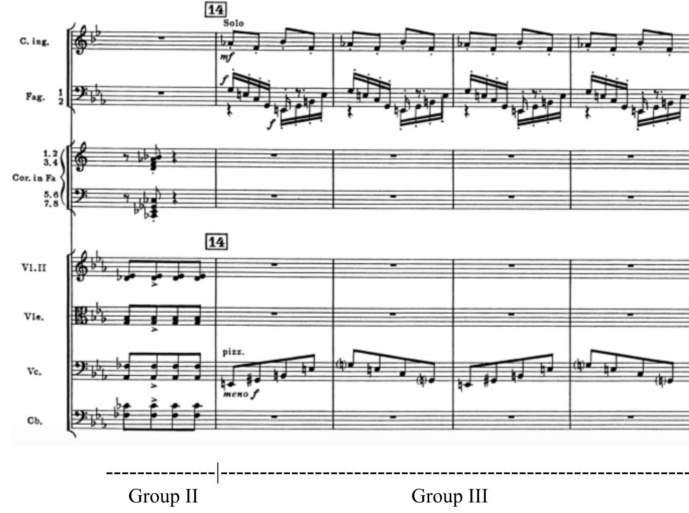
¹³ “Divertimento N15”, composed by Mozart, choreographed by Balanchine (1986, NYC Ballet), <https://www.youtube.com/watch?v=OD49rdHnYh8>.

¹⁴ “Bolero”, composed by Maurice Ravel and choreographed by Maurice Béjart, featuring Maya Plisetskaya and the Ballet du XXe Siècle (Bruxelles, 1975), <https://www.youtube.com/watch?v=SsSALaDJuN4>.

¹⁵ NYC Ballet’s Emilie Gerrity on George Balanchine’s DIVERTIMENTO NO.15, <https://www.youtube.com/watch?v=Bn7X2LP5bN0>



(a) In the dance, the grouping boundary between Groups II and III is marked by solo dancer detaching herself from the group who continues to dance the same dance pattern as in Group II.



(b) In the music, the grouping boundary between Groups II and III is marked by the change in character and instrumentation from an intense, rhythmic, bright, and dissonant ostinato (Group II), to a few low-pitched instruments playing a less contrasted and more fluid tune (Group III).

Figure 3: Bausch’s “Rite of Spring”, *Augurs of Spring* - “string ostinato section”: Groups II (4:29, fn.12) and III (4:38) are defined based on Charnavel’s (2023) syntactic grouping analyses. (Figures taken from Charnavel, 2023, Figs.19 and 4).

Intuition: Coherent.

Syntax: Congruence of music and dance grouping boundary between Groups II and III; incongruence between the group of dancers in the background and the music in terms of grouping (no phrase boundary between Groups II and III for the group of dancers) and meter (during Group III).



Figure 4: Balanchine’s “Divertimento N15” (12:08, fn.13; also see Table 2 of Krumhansl and Schenck, 1997): while the front ballerina finishes her solo and is about to leave the stage, another ballerina places herself at the back of the stage.

Intuition: Coherent.

Syntax: Congruent grouping boundary between two larger musical phrases and the switch of dancers.

candidate to explain our intuitive sense of dance-music coherence. For the syntactic hypothesis to hold, we must now verify that syntactic incongruence indeed induces incoherence.



Figure 5: **Béjart’s “Bolero”** (repetitions; 2:41, fn.14): The music is composed of one melodic theme repeated throughout the piece. This particular repeated dance motif reappears at the end of each full version of the musical theme.

Intuition: Coherent.

Syntax: Congruent.

3.1.2 Incongruence and Perceptual Effects

Incongruence arises when one or more structural parameters that are comparable between modalities do not map to each other. Intuitively, the hypothesis is that syntactic incongruence is perceived as dance-music incoherence – not “fitting” together. Merce Cunningham, for example, is known for choreographies where dance and music are deliberately incongruent, with the music acting as a sonic backdrop rather than as a rhythmic base (Fogelsanger and Afanador, 2006). For instance, “Second Hand” (2) was choreographed and composed independently, only to be paired for the first time at the very end. The result is clearly incongruent and rather abstract, in line with 20th century contemporary, more conceptual, dance (Fogelsanger and Afanador, 2006), and intuitively perceived as incoherent.

(2) **Cunningham’s “Second Hand”¹⁶:**

Dance and music each have structural characteristics that are comparable (identifiable tempo, rhythm, grouping boundaries, accents, etc.), but that do not map to each other.

Intuition: Incoherent.

Syntax: Incongruent¹⁷

More specifically, we can identify some particularly salient structural characteristics that are involved in generating incongruence:

Incongruence of accents. In Balanchine’s “Divertimento N15”, some jumps and *portés* (i.e. accented movements) aren’t synchronized to any comparably salient musical event (e.g. at 24:27, fn.13), compared to other highly congruent accents; this can be perceived as a breach to coherence.

Incongruence of rhythm (grouping, meter, tempo). Similarly to congruence, syntactic incongruence can be analyzed at a rhythmic level (grouping and meter), for which mismatches were formally described by Charnavel (2023). An example of rhythmic incongruence can be found between the background group of dancers and the music in Group III of *Augurs of Spring* (Bausch’s “Rite of Spring”, Fig.3). Tempo-wise, in Naharin’s “Venezuela” (1), when dancers skip synchronously and regularly across the stage, inducing a clear tempo, the concurrent music in Part 1 is fluid with no identifiable tempo. Plus, clear section boundaries in the dance do not map at all to that of the music in Part 1 (which are sparse and unclear). These rhythmical incongruencies do indeed induce incoherence.

One could therefore intuitively hypothesize that dance and music feel incoherent when they are syntactically incongruent. Incompatibility between dance and music syntax can indeed result in ambiguity

¹⁶For an extract from “Second Hand” (1970), composed by John Cage, choreographed by Merce Cunningham, see <https://www.youtube.com/watch?v=-FwiMIDQ7rI>.

¹⁷Exceptions include instances of capture giving the illusion of congruence.

between competing structures, in turn inducing uneasiness or instability (Charnavel, 2023). Dance and music may otherwise appear to simply be juxtaposed, with no points of contact, and if we were to only consider the congruent information between dance and music, we may have nothing left to interpret¹⁸. Then, would all incongruent dance-music sequences be perceived as incoherent?

Congruence Biases

Mitchell and Gallaher (2001) observed that swapped (incongruent) dance-music combinations could still be perceived as matching, so incongruence must be overcome in some way. Indeed, perceivers may be biased to prefer congruent syntactic information when forming a combined representation so as to reduce cognitive effort, as suggested by experimental evidence for a “capture effect” – the approximation of simultaneity allowing for slight (even perceivable) temporal asynchrony (Fogelsanger and Afanador, 2006; Watanabe and Shimojo, 2005). Being redundant, congruent features may also appear as more salient, reinforcing their representation in the overall perceived structure (Alais et al., 2010; Cohen, 2016)¹⁹. This stands out in the way music and dance emphasize and disambiguate each other’s structures (Charnavel, 2023; Howlin et al., 2017). If congruence does lead to coherence, then the illusion of congruence would allow for some incongruence without affecting coherence and thus make it hard for choreographers to achieve incoherence (Fogelsanger and Afanador, 2006). This could explain the arbitrary congruencies perceived in Howlin et al.’s (2017) and Mitchell and Gallaher’s (2001) studies.

3.2 Limits of the Syntactic Analysis

Despite the explanatory power of Theory 1, there are choreographical examples that point out shortcomings of a purely syntactic analysis.

Incongruent but still coherent: the forgotten role of incongruence

Crucially, the syntactic theory fails to adequately address incongruence, and biases towards optimizing for congruence don’t erase all incongruence either. While most studies have primarily focused on congruence (Fogelsanger and Afanador, 2006), a dance-music analysis cannot ignore incongruence: it is pervasive in choreographies, even in the most congruent ones.

If it were only for syntactic correspondences, incongruent sequences would be incoherent – strange, uninterpretable, or dismissed as irrelevant. However, there are audio-visual combinations in which incongruence is interpretable (Bolivar et al., 1994; Fogelsanger and Afanador, 2006; Howlin et al., 2017; Mitchell and Gallaher, 2001): there is something beyond structural features determining perception of coherence. As a basic example, it often happens that music is incongruently paired with stillness in the dance (and conversely). At some point in Bausch’s “Rite of Spring”, stillness in the dance is accompanied by loud intense music (Fig.6), but the lack of dance-music syntactic mapping doesn’t necessarily feel incoherent. It is surprising, but nonetheless comprehensible. Rather, when considering semantics, syntactic incongruence could have a meaning in itself (e.g. representing a dramatic situation). Otherwise, why would such incongruence be so common, even in Balanchine’s “Divertimento N15” or narrative dance such as Khan’s “Jungle Book Reimagined”²⁰, if it were to be interpreted as incoherent?

¹⁸We base ourselves on the presence/absence of congruence but do not yet seek to quantify ratios of congruence to incongruence or relative weight of features. Nonetheless, it arguably never happens that every single parameter is strictly congruent or incongruent (Godøy et al., 2016).

¹⁹Besides, slight asynchronies may also be dismissed as mere performance or film rendering errors, with the assumption that it was probably “meant to be” congruent in the first place.

²⁰ “Jungle Book Reimagined” is a modern version of Mowgli’s story turned into a choreography. Although it mixes dance, music, spoken text, and projected visuals, the dance-music relationships are core drivers of the story. A trailer can be found here: <https://www.youtube.com/watch?v=H6tjZfNRyFM>.



(a) After falling, the dancers lie still on the ground.

(b) At the same time at [53], the music is *fortissimo*, with *pesante* accented dissonant chords across the whole orchestra.

Figure 6: Bausch’s “Rite of Spring”, “Forte-Still” example (11:33, fn.12): The dancers suddenly fall to the ground and lie motionless while the music is *fortissimo*, accented, and dissonant.

Intuition: Coherent; dramatic effect surpassing the sum of that of each modality individually.

Syntax: Incongruent.

Semantics: Dance and music are individually comprehensible (stillness for dance, intensity and tension for music); combined, they are also comprehensible (e.g. climatic tension between the entities each modality represents).

This extends to other types of incongruence. In Cunningham’s “Beach Birds” (Fig.2), Cage’s music (rain-sticks) is clearly structurally incongruent with the dance. Accordingly, Cunningham intended the

phrasing of the dance movements to be fluid, unsynchronized, and of a variable length. In spite of this successful structural inconsistency, it doesn't feel as incoherent as syntax would predict. In fact, it's even coherent: one can easily interpret it as representing a scene with birds by a seashore (Fig.2b). What tool could account for coherence if syntactic congruence cannot? Cunningham indeed meant this choreography to imitate a flock of birds on a beach²¹ so there are reasons to think that a semantic dimension may be at play in the overall sense of coherence perceived.

Congruent but still incoherent

Do all syntactically congruent dance-music combinations feel coherent? It seems unlikely. If the fictional "Angry-Happy" example (Fig.1) feels incoherent – even when the dancer's steps and change of direction are syntactically congruent with the music, then, assuredly, it has to do with the tight fists and stomping feet that contrast with the joyfulness of the music – an expressivity that is inherently semantic.

Nuancing syntax

There are internal nuances to coherence that syntax may be overlooking. Firstly, syntax is limited by the contiguity principle, whereas dance-music relations can transfer beyond contiguous segments (Schlenker, 2022). In Béjart's "Bolero" (Fig.5), each repeated instance of the same dance movements appearing at the end of each melodic iteration can be naturally bound together, yet these associations overpass constituency. This non-contiguity problem occurs also without repetitions. For example, in Bausch's "Rite of Spring" (*Ritual of the two rival tribes*, 13:08, fn.12; see Appendix for a more detailed analysis), male and female dancers form two defined groups on stage, and are associated with two main musical motifs that alternate accordingly. These systematic but non-contiguous associations determine which group dominates the stage. Such transfers would only be possible if there were variables indexing dance-music relations across time (i.e. semantic interpretation of discourse referents, such as two rival tribes alternating their actions).

Secondly, semantics can go beyond congruence biases, better accounting for some intuitive judgments of coherence. For instance, coherence in the "gunshot" example from Khan's "Jungle Book Reimagined" (3) is driven by the semantic relationship between the music (gunshot-like) and the dance (fall) events rather than by capture (that would make them congruent). The syntactic theory fails to account for interpretations of music and dance having distinct roles in the overall story.

(3) Simplified from Khan's "Jungle Book Reimagined" ("gunshot" example):

A dancer suddenly falls to the ground slightly after a loud sudden gunshot-like sound in the music.

Intuition: Coherent as 2 separate events related by a causality relationship.

Syntax: Incongruence but possibly resolved as congruence by the capture effect.

Semantics: Comprehensible (2 denoted objects, the action of one causing the other to fall).

Reassessing Theory 1 (syntax only) in light of our observations, it appears that a strictly syntactic analysis may not be the only tool to understand dance-music interactions. Firstly, syntax fails to draw proper distinctions between cases of incongruence, which are not necessarily incoherent. Plus, it turns out that many syntactic concepts associated with incongruence (i.e. tension, instability, ambiguity) are also analyzable semantically, so incongruence can have a meaning in itself. Secondly, syntactic congruence doesn't properly distinguish the different kinds of coherent dance-music relationships, and most importantly congruence can also be incoherent. Thirdly, because it is comparison-based, syntax is

²¹See "Beach Birds - Merce Cunningham Trust — mercecunningham.org", n.d.

limited to dance-music juxtaposition and thus doesn't provide the tools necessary to understand if and how dance and music combine into a whole single unified percept.

4 Theory 2: Adding a Semantic Layer to the Analysis of Music-Dance Sequences

In light of Theory 1's shortcomings, we must extend our theory: we hypothesize that, in addition to the syntactic theory, a semantic analysis of dance-music sequences is not only possible but also a necessary tool for dance-music analysis. In fact, (i) the examples analyzed so far suggest that multimodal relations do exist at a semantic level; (ii) there exist semantic theories of dance and of music that we can start from; (iii) dance and music semantics are abstract enough to bypass modality differences. More concretely, could examples such as those presented in Section 3.2 be better explained by a semantic analysis? If not, then a semantic level of analysis would not be necessary to account for dance-music coherence intuitions.

Section 4.1 will present the building blocks of formal semantic methodology applied to audio-visual multi-media (Schlenker, 2022) and how they can extend to dance-music sequences. I then propose two ways by which dance and music combine their meanings in a holistic way, using well-established semantic concepts: conjunction (Section 4.2) and discourse referents (Section 4.3). Reevaluating the issues raised in Section 3.2, Section 4.4 compares predictions of coherence judgments between the combined semantic analysis (comprehensibility) and the syntactic analysis (congruence).

4.1 Truth Beyond Language: The Semantics of Music and Dance

The goal of this endeavor is to show that a semantics of non-linguistic sequences shows non-random patterns going from stimulus features to inferences, and can thus be analyzed systematically. One such way is by applying truth-conditional semantics to non-standard objects: a sequence is true relative to a situation if it satisfies certain truth (satisfaction) conditions (Patel-Grosz, Grosz, et al., 2022; Schlenker, 2022). Just as "it rains" is true of all situations in which the truth condition "it is raining" holds but is false if it isn't raining, a crescendo in music can be true of all situations in which the following truth conditions hold: either there is a gain in energy or an object is moving closer (Schlenker, 2022).

Satisfaction conditions are comparable between dance and music. For example, *silence* in music may indicate that an event is interrupted (Schlenker, 2018) which can parallel the dance parameter *stillness*. When modalities combine, meaning can typically be derived from resemblance (i.e. iconicity) between a dance-music event and an audio-visual external situation, which coherence may be important for intuitive judgments. Emotion expression is also a form of meaning relevant to both dance and music that seems at play in modulating dance-music coherence (e.g. "Angry-Happy" example, Fig.1).

Another contribution from formal semantics is the formulation of meaning with the use of discourse referents (i.e. variables) to which semantic assignment functions assign objects²². Recent investigations showed that relations of coreference and disjoint reference among discourse referents were also relevant to music, especially paired to images (Schlenker, 2022), and to narrative dance (Patel-Grosz, Grosz, et al., 2022; Patel-Grosz, Katz, et al., 2022).

²²These denoted objects can be of various natures (e.g. a bird, a geometrical shape, a narrator) as they are underspecified by the stimulus.

Semantics of mixed sequences

To better understand the semantics of mixed sequences, we start from simple examples of combined video and sound as concrete probes for audio-visual coherence intuitions, because they trigger clearer semantic inferences, while still exhibiting similar patterns of meaning formation (Schlenker, 2022) – with additional refinements needed to account for choreographies’ own complexities.

Schlenker (2022) takes a cartoon-music example from the “Sorcerer’s Apprentice” (the beginning Disney’s *Fantasia*, 1940) to illustrate semantic interpretations of discourse referents of mixed sequences (Fig.7). He shows how discourse referents display semantic cross-referencing between modalities, such as between specific sections of the music and characters in the images. It turns out that these associations influence the possible interpretations of discourse referents of the music itself, in which two otherwise similar sections (suggesting coreference) in fact represent disjoint referents (see example (59) of Schlenker, 2022). This demonstrates how audio-visual combinations favor certain interpretations of discourse referents within modalities which would’ve been more ambiguous on their own, by “maximiz[ing] correspondence between the two media” (Schlenker, 2022)²³.



Figure 7: “Sorcerer’s Apprentice” (Disney’s *Fantasia*, 1940): Example of discourse referents assignment for four pictures (below) and their corresponding musical events (above), with v_1 representing the sorcerer, v_2 the apprentice, and v_3 the genie (taken from Schlenker, 2022, (60)).

Cross-referencing of discourse referents also appears in combined music and dance. In Schlenker’s (2022) analysis applied to Fokine’s ballet “Les Sylphides”²⁴, he suggests that discourse referents represented by specific sections in the orchestration map to that of the main ballerina while variations of these sections correspond to the other dancers in a way that influences which relations among discourse referents are possible. Indeed, through the creation of dance with music, there are decisions made as to which relations between discourse referents should stand out by manipulating patterns of cross-modal cross-referencing.

Moreover, Schlenker (2022) proposes that the semantic interpretation of joint music and visual animations is done through the conjunction of their semantics. Most importantly, this approach can account for complementarity between modalities; each contributes to the general meaning by licensing different semantic information without just being representations of each other. Clair (1953, as cited in Fogelsanger and Afanador, 2006) provides a telling audio-visual example – “a car door slamming shut needs only be heard but not seen” – to show how sound can give information about a narrative without it being shown visually. Similarly, dance and music have their own – complementary – abstract semantics that do have points of contact but need not convey redundant information to be comprehensible (Schlenker, 2022).

²³This also involves patterns of coindexation among discourse referents that counter syntactic constraints of constituency relations (contiguity).

²⁴Fokine’s “Les Sylphides” – originally called “Chopiniana” (1984, American Ballet Theatre), ballet choreographed on Chopin’s Mazurka Op. 33 No 250, on an orchestration close to Britten’s version, <https://youtu.be/Lsuc3KUKKpQ?feature=shared> (as cited in Schlenker, 2022).

Mixed-sequence semantics and coherence

The case of speech with gestures (Lascarides and Stone, 2009) suggests that coherence is intricately linked to mixed sequence semantic comprehensibility. As for incoherence, it may arise from a form of semantic contradiction between modalities, in sequences for which “no set of appropriate parameters and arguments satisfies it” (Schlenker, 2022). A sequence that is true of no situation renders incomprehensibility (e.g. a bomb exploding concurrently with a stable chord in the music).

To gain concrete intuitions about the coherence of mixed sequence semantics relative to a syntactic theory, consider a video clip with an explosion as the visual event and an explosive “boom” as the auditory event. By varying the synchronization of the sound to the visuals, the number of sounds or its timbre, we can differentially modulate structural and semantic audio-visual relationships, and compare them to intuitions of coherence. The five example combinations in (4, illustrated in Fig.8) set the stage for our exploration of dance-music combinations throughout Section 4. Crucially, it reveals a pattern: semantic comprehensibility seems to be what determines perceptual coherence, rather than syntax.

(4) **Audio-Visual “Explosion” Example** (see Fig.8 for a visual illustration):

- a. A loud intense “boom” noise of an explosion synchronized with the visual explosion feels intuitive (unless the explosion is in space) and would realistically represent a situation in which there is an explosion – a startling but coherent inference.

Intuition: Coherent.

Syntax: Congruent (synchronized).

Semantics: Comprehensible, 1 denoted object (1 explosion).

- b. The sound of the explosion comes 3 seconds after the visual explosion. A coherent inference is that of an explosion taking place far away (approximately 1km). In comparison, this clip would not be true of a situation in which there are two explosions (with the 2nd being out of sight), but an explosion that doesn’t make any sound would be semantically incomprehensible.

Intuition: Coherent.

Syntax: Incongruent (unsynchronized, Visual < Audio).

Semantics: Comprehensible, 1 denoted object (1 explosion).

- c. The sound of the explosion takes place 3 seconds before the visual explosion. Here, it is intuitively incoherent; this clip is true of no situation.

Intuition: Incoherent.

Syntax: Incongruent (unsynchronized, Audio < Visual).

Semantics: Incomprehensible.

- d. The sound of the explosion appears twice: 3 seconds before the visual explosion (as in 4c), then synchronized with the visual explosion (as in 4a). Even if it includes (4c) which is incoherent, this version would still be semantically comprehensible and coherent: a possible inference is that there are two different explosions, with the first one being out of sight.

Intuition: Coherent.

Syntax: Incongruent (unsynchronized, Audio < Visual); Congruent (synchronized).

Semantics: Comprehensible, 2 denoted objects (2 explosions).

- e. If a sound is synchronized to the visual explosion, but this time it sounds like a drop softly falling in a glass of water. What would the possible inferences be? Even with a congruent syntax, a semantic analysis (no comprehensible denotation) predicts incoherence.

Intuition: Incoherent.

Syntax: Congruent (synchronized).

Semantics: Incomprehensible.

Combination	Temporal sequence of events			Intuition	Syntax	Semantics
	-3s	0	+3s			
a				Coherent	Congruent	Comprehensible
b				Coherent	Incongruent	Comprehensible
c				Incoherent	Incongruent	Incomprehensible
d		 		Coherent	Incongruent + Congruent	Comprehensible
e		 		Incoherent	Congruent	Incomprehensible

 Video of explosion (visuals only)

 Sound of explosion

 Sound of drop falling in water

Figure 8: **Audio-visual “explosion” example:** Visual illustration of each audio-visual combination described in (4). The visual sequence (in pink) is vertically synchronized to the auditory sequence (in blue), centered at the visual explosion at $t = 0$.

Towards principles of a dance-music semantics

When adding dance to music, viewers cannot be expected to infer the ‘correct’ denotations the choreographer/composer intended to communicate. Yet, viewers are constrained in terms of which denotations they can infer – and even more so than with music alone or dance alone. So, what kind of framework would be most appropriate to approach how dance and music denotations combine, and does it necessarily require a combined semantic theory?

4.2 Combining Sets of Denotations: Conjunction

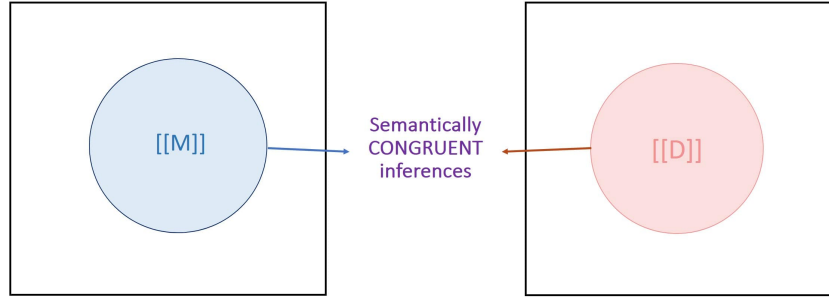
Previous investigations suggest that dance and music have their own semantic inferences, but fall short of an integrated semantics of combined dance and music. Taking as input the sets of denotations of music $[[M]]$, and of dance $[[D]]$, we will consider three hypotheses to evaluate their resulting dance-music semantics $[[S=<M,D>]]$ (Fig.9): (1) semantic congruence (similarity between each modality’s semantics), (2) union (additivity), and (3) conjunction of each modality’s set of denotations.

Hypothesis 1: Semantic Congruence

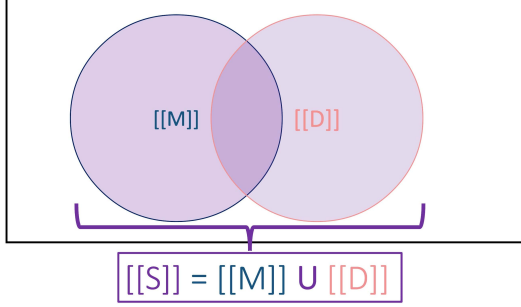
A semantic congruence analysis (Fig.9a) of combined dance and music consists in computing the degree of similarity between their truth conditions as semantically congruent (similar) or incongruent (dissimilar)²⁵. This does not imply the necessity for a combined semantics of dance-music sequences and only involves semantic parameters that can be compared (e.g. energy level, speed, emotions).

As a simple example, take a music sequence M that stops into silence and a dance sequence D that stops into stillness. Since both M and D are true of situations with interrupted movement, their truth

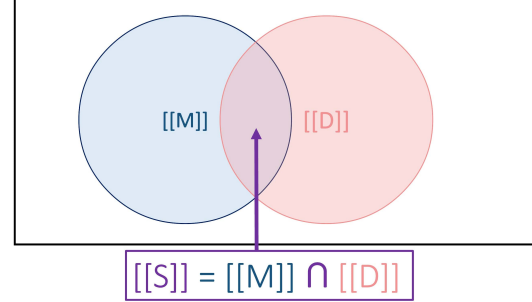
²⁵“Similarity” between such underspecified sets of denotations may however be hard to define.



(a) **Hypothesis 1:** Dance and music semantics are juxtaposed. Possible inferences of a sequence are limited to meanings that are congruent between dance and music. This hypothesis accounts for dance-music meaning correspondences but fails to account for comprehensibility when there is semantic incongruence.



(b) **Hypothesis 2:** Dance and music meanings add up. The set of denotations that a given dance-music sequence is true of is the union of the individual sets of denotations. This hypothesis accounts for comprehensibility with semantic incongruence but wrongly predicts truth of denotations that aren't represented by a joint sequence.



(c) **Hypothesis 3:** The set of denotations that a given dance-music sequence is true of is the conjunction (intersection) of the individual sets of denotations. This is the most plausible predictor of dance-music meaning and comprehensibility.

Figure 9: Three hypotheses as to how joint dance and music form a dance-music semantics.

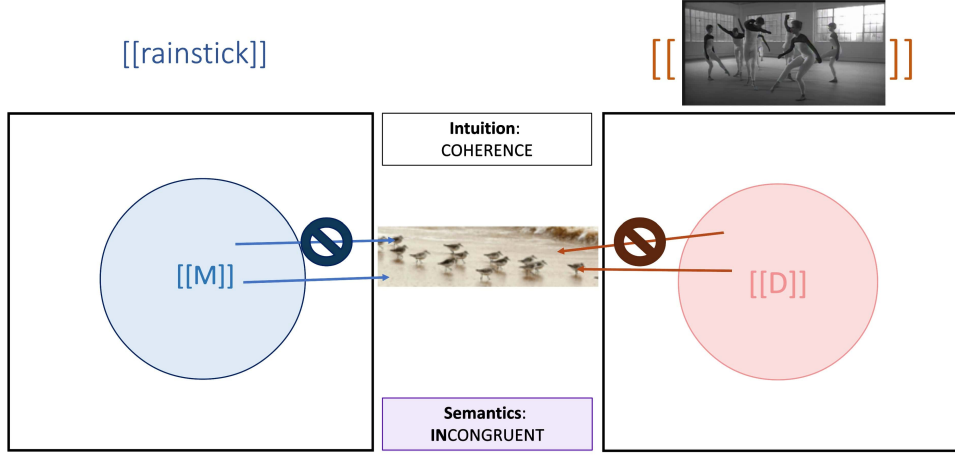
conditions are congruent. Now, if another music sequence M' continues and is paired with D that stops, then they are semantically incongruent since their movement property is opposite.

Dance-music semantics by congruence is limited as it cannot account for coherence of sequences with non-similar denotations. It would therefore not account for Cunningham's "Beach Birds" (Fig.2), in which each modality represents something different (Fig.10a) but still appears to be comprehensible. Identifying semantic correspondences between dance and music is not sufficient to understand dance-music integration as it fails to capture complementary information that dance and music each bring to the general percept: a semantic analysis of mixed sequences must also address semantic incongruence (Schlenker, 2022).

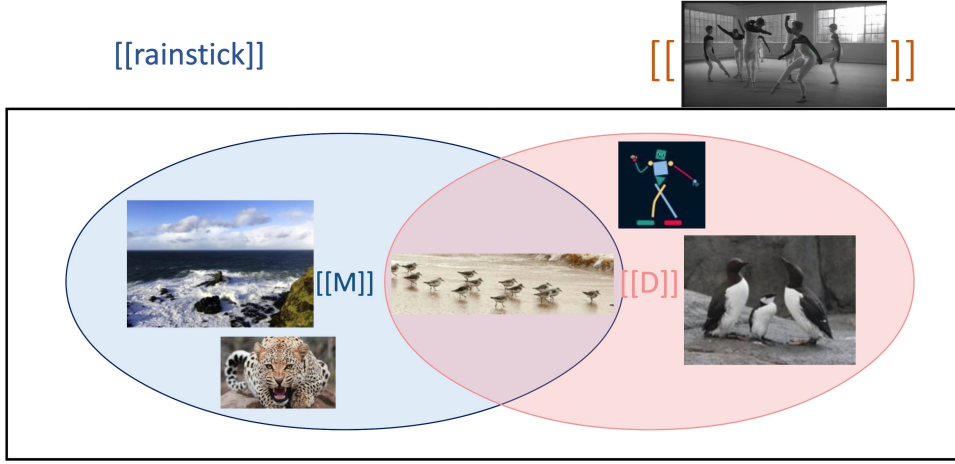
Hypothesis 2: Union

To account for complementarity and incongruence, one could hypothesize that dance and music semantics add up to each other through union of their sets (Fig.9b). This hypothesis is inspired from Krumhansl and Schenck's (1997) experimental results. In their study, they measured separate and combined effects of watching the music and dance of an extract from Balanchine's "Divertimento No.15" (fn.13). Using a multiple regression model, they demonstrated that structural features, as well as continuous ratings of tension and emotion expression of the combined stimuli could be computed as the sum of that of each individual modality's ratings. In short, dance and music would each contribute to the combined percept without influencing each other.

Testing this hypothesis with our "Beach Birds" example, union rightfully considers it comprehensible



(a) The music may denote *water* but not *birds*. Conversely, the dance may denote *birds* but not *water*. Semantic congruence would therefore predict that this dance-music sequence is not true of the situation depicted in the picture. This is inconsistent with intuitions that this sequence is true of situations with birds on a seashore.



(b) The music may denote situations with *water* or with a *growling tiger*. The dance may denote situations with *birds* or even *robots walking awkwardly*. The union hypothesis predicts that any situation that the dance is true of or that the music is true of are also true of the joint sequence (e.g. *birds on a seashore*, but also *a robot walking awkwardly*, or a *water-shore without birds*). The conjunction hypothesis predicts that only situations that both the dance and the music are true of (e.g. *birds on a seashore*) are true of the joint sequence.

Figure 10: **Semantic analysis of Cunningham’s “Beach Birds”** (music composed of rain-sticks and dance of awkward contrasted movements; Fig.2): It is a semantically incongruent but comprehensible dance-music sequence when analyzed by union or conjunction.

(Fig.10b), but also erroneously predicts that the combined sequence also denotes situations of which the dance or the music are not true (e.g. *a seashore without birds*). Krumhansl and Schenck’s (1997) approach is therefore too narrow and their results unlikely to generalize beyond Balanchine’s highly congruent works. Moreover, the fatal flaw of the Union hypothesis is to predict that contradiction is coherent (e.g. “Angry-Happy” example, Fig.1). This is thus unlikely to be an appropriate account of dance-music semantics²⁶.

²⁶Note that if we could simply evaluate whether a situation σ is true of a dance-music sequence by checking if σ is in the set of denotations of at least one of the modalities, then the three different choreographies analyzed by Charnavel (2023) on Stravinsky’s “Rite of Spring” would be equally true of the music’s denotations and similarly comprehensible.

Hypothesis 3: Conjunction

Taking from a well-established semantic concept and Schlenker’s (2022) suggestion, I now hypothesize that a combined semantic analysis of dance-music sequences takes as input the set of denotations of the dance and music sequences, and outputs their intersection (i.e. conjunction) as a single set of denotations true of that dance-music sequence (Fig.9c). That is, a joint sequence S is true of situations σ within $\{[[M]] \cap [[D]]\}$ ²⁷.

Meaning in Semantic Incongruence

Taking Cunningham’s “Beach Birds” example again, this joint sequence is comprehensible – and indeed intuitively coherent – when considering the denotations that both modalities are true of (e.g. *birds with water*), within the conjunct set of their denotations (Fig.10b). This example illustrates how conjunction is able to extract meaning when there is semantic incongruence while preserving overall coherence without over-estimating the inferences this sequence is true of. Conjunction makes complementarity of dance and music meaning possible, as long as inferences (1) contain a representation of the dance, (2) contain a representation of the music, and (3) are semantically understandable (these representations are compatible within a coherent whole). The conjunction hypothesis thus offers an alternative explanation to previous findings such as Mitchell and Gallaher’s (2001) surprising results in which participants shown incongruent dance and music pairs agreed with the statement that they matched.

Incoherence

Conjunction also predicts incoherence. In the “Angry-Happy” example (Fig.1), the conjunction analysis overcomes the union hypothesis’ issue of over-predicting coherence in cases of incompatible incongruence. Even if each modality is individually comprehensible, their combination feels incompatible. The intersection of the dance’s and music’s sets of denotations would theoretically be null, yielding an incomprehensible – and indeed incoherent – pair. Now, keep the same music and simply replace the stomping of the dancer by skipping and untighten their clenched fists: we have not changed anything syntactically, but this time there is considerable overlap in their sets of denotations. This new dance-music combination is semantically comprehensible by conjunction – and is indeed coherent.

One dance, two different musics: two distinct semantics

Recall Naharin’s “Venezuela” (1): conjunction would predict that two different music sequences with the same dance will yield completely different sets of meanings (Fig.11)²⁸. Indeed, subjectively, the two parts feel clearly different on a semantic level (relationships between dancers, emotions, general story), as if the sets of denotations of each combination were distinct.

In whole, compared to the semantic congruence and union hypotheses, conjunction reflects finer, richer forms of information of dance-music combinations by joining their meanings into one single scene and accounting for semantic in/congruence, complementarity, and incoherence. It provides a useful tool to evaluate truth conditions and semantic acceptability that seem consistent with intuitive coherence. It is

²⁷Two border cases constitute limitations to the theory: (a) If dance and music denotations are the same ($[[S]] = [[M]] = [[D]]$), then the sets of denotations perfectly overlap. This is very unlikely due to obvious modality differences intrinsically providing different types of semantic information. (b) If dance and music denotations are mutually exclusive, then no denotations are possible ($[[S]] = \emptyset$) – which clearly renders incomprehensibility. Such a clear-cut case is unlikely due to under-specification and cognitive biases towards maximizing points of contact. The closest example we have seen is the “Angry-Happy” example (Fig.1).

²⁸This is the prediction of a simple conjunction theory – a completely null intersection for Part 1 may be impossible in practice. In a more balanced theory, the dance-music intersection for Part 1 would be relatively smaller than that for Part 2.

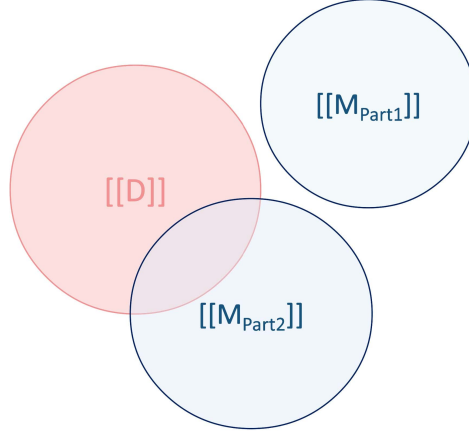


Figure 11: Simplified schematic representation of a conjunction analysis of Naharin’s “Venezuela” (1). The dance doesn’t appear to share common denotations with the music from Part 1, but it does share denotations with the music from Part 2. Whereas the former dance-music pair is deemed incomprehensible, the latter pair is semantically acceptable.

functional on its own, but remains quite abstract. Taking a different angle, the following section proposes a more fine-grained analysis of comprehensibility and coherence at the level of denoted objects.

4.3 Combining Discourse Referents

A discourse referents analysis, inspired from Schlenker (2022) and Patel-Grosz et al. (2022), supplies crucial tools for understanding dance-music inferences. In semantics, discourse referents are variables to which are assigned objects, so as to distinguish referents from one another and refer back to them later. The present claim is that each modality’s discourse referents can combine into a common dance-music semantics by pointing to (i) a same object in a same external reality simultaneously or at different points in time (i.e. coreference; Section 4.3.1), or (ii) distinct objects in a same external reality (i.e. disjoint reference; Section 4.3.2).

4.3.1 Dance-Music Coreference

Multimodal coreference corresponds to the coindexation of discourse referents from different modalities so as to reference a single object in a combined semantic analysis. Just as redundant information from a visual and an auditory explosion (4a, 4b, Fig.8) leads to coindexation of discourse referents from distinct modalities into one denoted object, discourse referents from dance and music can coindex a same object. This can arise from cross-reference, as in Fokine’s “Les Sylphides” (see Section 4.1) or Balanchine’s “Divertimento N15” in which discourse referents from sections of the music may be coreferencing with that of specific dancers (as in Fig.4).

Coreference relations may later serve to evoke the same object even if only one of the modalities reproduces the associated sequence. This reflects meaning inheritance across modalities and across time through assignment of discourse referents which either modality can later refer to. This strategy is extensively used in Khan’s “Jungle Book Reimagined” (fn.20) to drive the coherence of the narrative. Coreference can also explain examples from Section 3.2 in which dance-music associations cross syntactic constituency barriers (*Ritual of the Rival Tribes* from Bausch’s “Rite of Spring”, also see Appendix; Béjart’s “Boléro”, Fig.5).

Summarizing, by showing how a single object can be denoted by multiple modalities, coreference sheds light on the fusion of modalities into one holistic semantic representation. The total number of

denoted objects is also reduced compared to the total sum of the discourse referents in the dance and music. This counteracts underspecification of each individual modality, as objects inherit satisfaction conditions from two sources of information instead of one. Rather than modalities being just juxtaposed, coreference strengthens dance-music bonding, which may in turn drive coherence.

4.3.2 Dance-Music Disjoint Reference

Coreference alone is too narrow to account for dance-music complementarity. It must be supplemented by dance-music disjoint reference, which can take place when a joint sequence is true of situations in which discourse referents (from distinct modalities) reference distinct objects. Importantly, it captures how dance- and music-denoted objects can coexist in a combined, single, comprehensible choreography-external reality, rather than denoting their own, independent, modality-specific external realities.

Disjoint reference is possible even when coreference also satisfies a sequence, provided this inference is semantically acceptable. This implies that multiple possible patterns of relations between discourse referents may be possible for a given dance-music sequence. In the example from Béjart’s “Boléro” where similar movements are repeated at the end each version of the theme (Fig.5), dance-music coreference could have been possible but a disjoint relation is comprehensible as well, with a dance discourse referent and a music discourse referent representing two different objects that happen to come together at regular points in time. It turns out that some disjoint reference interpretations even feel more intuitive than their coreference counterparts, especially in more narrative choreographies such as Khan’s “Jungle Book Reimagined”. In the “gunshot” example (3), the accented fall of the dancer on the ground and the gunshot-like accent in the music could possibly corefer to the same object (such as the musical accent being the sound of the dance object falling). Yet, disjoint reference is a more natural inference, with the music-denoted object causing the dance-denoted object to fall (e.g. a bullet hitting an animal). This shows that multimodal semantic relations are powerful enough to over-pass congruence biases.

Moreover, disjoint reference can account for coherence of dance-music sequences which would have been evaluated as semantically incomprehensible if it were only for cross-reference or coreference, provided their respective denoted objects can coexist in a single comprehensible external reality when disjoint. This is how the dramatic *fortissimo* music combined with dancers lying still on the floor from Bausch’s “Rite of Spring” (Fig.6) can be deemed semantically comprehensible. By interpreting the dance and music as representing distinct objects with different motion and/or energy states within the same situation, this opens up possible denotations such as the music representing an external narrator (or a punishing pagan god) and dance representing a crowd. Similarly, Cunningham’s “Beach Birds” (Fig.2) is true of situations in which each modality’s denoted objects are compatible when disjoint (e.g. *birds, a seashore* as in Fig.2b), perhaps explaining its coherence.

Multiple discourse referents

How could semantics contribute to predicting coherence of more complex dance-music sequences in which dance and music each represent several discourse referents? Coreference and disjoint reference can operate in parallel to evoke more complex meanings. In the *Augurs of Spring* from Bausch’s “Rite of Spring” (Fig.3), the discourse referents analysis even sheds light on the possibility of musical occlusion through a mix of meaning inheritance and disjoint reference²⁹.

Although there may be multiple possible semantic assignments of discourse referents, not all are comprehensible. As an illustration, in a discourse referents analysis of an extract from Khan (Fig.12),

²⁹A more detailed analysis of the example can be found in the Appendix (Fig.13).

only three out of five interpretations of relations among discourse referents are semantically acceptable³⁰. Such patterns of interpretability highlight the productivity of a dance-music semantic analysis.

Incoherence

Are there sequences with no inferable discourse reference patterns? This is the case in the “Angry-Happy” example (Fig.1), for which it would be uncanny to have either one or two denoted characters in one same situation. Indeed, sometimes even disjoining dance and music objects feels incomprehensible: the power of dance-music disjoint reference is that it can predict incoherence.

Many interpretations of discourse reference relations may be possible, but possible inferences only include those that yield comprehensible scenarios. This semantic level of holistic multimodal processing captures more fine-grained complexities of the range of possible dance-music inferences in a way that is more representative of dance-music complementarity, compared to having two disjoint representations denoting two independent external realities - and this may be driving overall in/coherence.

4.4 Against a Purely Syntactic Analysis

A semantic analysis of combined dance and music formalizes rule-based constraints on possible inferences using conjunction and relations of discourse referents. Based on these principles, the goal of Section 4.4 is to compare syntactic and semantic predictions of coherence to empirically determine whether a semantic analysis of dance with music is actually necessary. We will first review syntactically incongruent (Section 4.4.1) and then syntactically congruent examples (Section 4.4.2) of dance-music pairs, showing how the examples introduced in Section 3.2 are in fact best accounted for by semantic comprehensibility.

4.4.1 Semantics Distinguishes Between Types of Syntactic Incongruence

Comprehensibility can predict coherence of syntactically incongruent sequences

Coherence can arise from meaningful relations among discourse referents, even in structurally incongruent sequences. This can be illustrated in the audio-visual domain by combination 4b of the “explosion” example (Fig.8) in which the visual and auditory explosions coreference one same explosion even though the sound plays 3 seconds after the visuals, while remaining coherent. Combination 4d (two explosion sounds, the first being incongruent – 3 seconds before the visuals, and the second being synchronized with the visual explosion) is also coherent and comprehensible regardless of its partial syntactic incongruence by disjointly referencing the early sound-denoted explosion (out of our visual field) and the second explosion.

In the choreographical domain, we have seen that contrary to the apparent incongruence (syntactic and semantic) of Cunningham’s “Beach Birds” (Fig.2), this music-dance combination is still coherent – and is in fact semantically comprehensible, true of its own specific set of situations. Indeed, using the conjunction hypothesis (Fig.10b), this sequence is only true of situations that are true of both modalities (e.g. *a group of beings with a stiff and clumsy demeanor near a rustling sound*). In terms of discourse referents, this sequence is true of comprehensible situations in which the dance and music objects are disjoint. It thus appears that, here, it is the combined, multimodal, semantic comprehensibility that makes the choreography coherent, not the syntax.

³⁰For simplicity’s sake, I have considered the music as having one discourse referent, but it could further be separated in two (singing and accompaniment). This would show more complex and interesting relations with the dance (e.g. the accompaniment coreferencing with the background dancers and the singing coreferencing with foreground solo dancer), but the argument would be similar: some interpretations of discourse referents are more comprehensible than others.

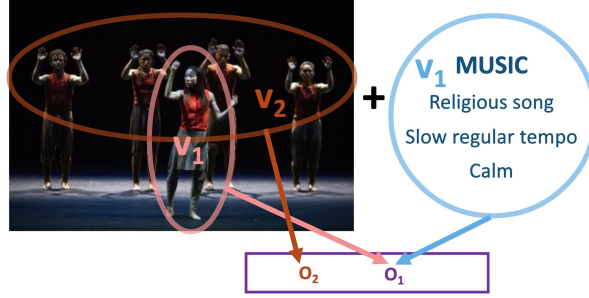


Figure 12: Khan’s “Jungle Book Reimagined” (fn.20) – possible semantic analysis of 3 discourse referents: There are 2 streams in the dance: a solo dancer in the foreground makes slow continuous movements, and a group of 4 dancers in the background hold a still posture. This co-occurs with 1 music stream: a gentle religious song with a calm instrumental accompaniment and a slow regular tempo (as in this track from Jocelyn Pook’s music for this choreography – <https://youtu.be/Xg2k0ePK2P4?feature=shared&t=75>).

Intuition: Coherent.

Syntax: Incongruent.

Semantics: Comprehensible. Considering each stream as a discourse referent, we can identify 5 combinations of coreference relations, 3 of which are comprehensible – *a*, *b*, and *c* – and 2 of which are incomprehensible – *d* and *e* (the joint sequence is not true of these situations).

In the “Forte-Still” example from Bausch’s “Rite of Spring” (Fig.6), recall that the music continues to rise in intensity while the dancers collapse to the ground, suddenly still. In terms of semantic interpretation of discourse referents, that their respective denoted objects are disjointly compatible within a same situation suggests possible coherence. Applying conjunction, this combination is true of inferences that involve a tense situation for example. The semantic acceptability of this sequence is thus more in line with our coherence judgment, whereas its incongruent syntax would have predicted incoherence. Interestingly, the discrepancy between dance and music semantics may actually have a crucial role by exacerbating the tension already communicated by the dissonant chords in the music. Incongruence can have a meaning in itself; overall combined semantics goes beyond that of each modality’s semantics individually, into an emergent merged meaning.

Furthermore, syntax can be forced to be interpreted as incongruent to preserve narrative coherence of a choreography. As we have seen in Section 4.3.2, Khan’s “gunshot” example (3) is most readily interpreted semantically as disjoint reference, thus counteracting the capture effect towards perceiving congruence between the musical accent (gunshot-like) and slightly delayed dance accent (fall), in turn influencing the syntactic interpretation as two distinct - actually incongruent - events.

Incomprehensibility can predict incoherence of syntactically incongruent sequences

In other cases of syntactic incongruence, dance and music do indeed feel incoherent, but they also happen to be semantically incomprehensible. This can be illustrated by the incoherence of a soft sound of a droplet falling in water taking place 3 seconds after a visual explosion (an incongruent hybrid version of combinations 4b and 4e of the “explosion” example, Fig.8). In Cunningham’s “Second Hand” (2), dance and music syntax were meant to be incongruent. It is unclear whether their semantics could combine within an external situation without being contradictory, but the lack of semantic relations between the dance and music renders it rather incomprehensible as a whole. The overall effect is indeed incoherence, as predicted by syntax (without capture)³¹.

In some clearer cases of incoherence, dance and of music are not only semantically incomprehensible but incompatible or even contradictory³². We can parallel this with combination 4c of the “explosion” example (Fig.8) in which the sound of the explosion only plays 3 seconds before the visual explosion. This incongruent audio-visual combination can be seen as contradictory, and is indeed incoherent: the visuals communicate that “there is an explosion” (implying that we should hear it), while concurrently, the audio communicates that “there is no explosion”. As a real-life example, recall the utter dance-music incongruence of Part 1 of Naharin’s “Venezuela” (1). The dance tends to be very rhythmic (clear tempo, accents, regular skipping, periodic sways, sharp contrasts). Semantically, it evokes periodicity, dynamism, and energy, but also sensuality, violence, and intensity. The music in Part 1, on the other hand, feels as if in contradiction: it lacks clear tempo, it is slow, has few accents, weak temporal organization, and few contrasts; it evokes a low energy, monotony, calmness, and a slightly negatively valenced religious atmosphere. The sharp contrast between dance and music is not just rhythmic, but also semantic (violence vs calmness, sensual vs religious, contrasted vs monotonous).

We therefore see that multimodal in/comprehensibility overtakes incongruence in predicting in/coherence. Most often, we don’t even notice dance-music structural incongruence when the story seems to “fit” to

³¹Nonetheless, the theoretical semantic compatibility between dance’s and music’s semantics by disjoining referents could nuance this intuition, possibly explaining some of Mitchell and Gallaher’s (2001) results.

³²Keep in mind that given the abstractness and underspecification of dance and music semantics along with our brain’s tendency to want to make sense out of things, semantic contradiction or true incoherence between dance and music is understandably hard or even impossible to induce (Fogelsanger and Afanador, 2006; Vincs et al., 2010). Nonetheless, even interpretations aimed to avoid contradiction (e.g. disjoint or background/foreground relationships), may remain unintuitive if dance and music inferences are unrelatable logically.

the point that semantic continuity can even modulate syntactic interpretations, the key being logical relations between each modality’s denotations. Instead of discarding it, semantics nuances incongruence by identifying possible interpretations. How does semantic comprehensibility fare at predicting in/coherence for syntactically congruent choreographical examples?

4.4.2 Semantics Distinguishes Between Types of Syntactic Congruence

Semantic incomprehensibility can predict incoherence of syntactically congruent sequences

Even in the audio-visual realm, syntactic congruence doesn’t imply semantic acceptability. For instance, in combination 4e of the “explosion” example (Fig.8), the visual explosion is congruent with the sound of a droplet falling into water, yet, these two unrelated events are semantically incomprehensible when put together – and indeed incoherent. A music-animation example of a visual explosion with a stable chord in the music would be contradictory as well (Schlenker, 2022). Similarly, a dance-music analogy would be a dancer stomping on stage, energetically hitting the floor, synchronized with violins playing pizzicato wouldn’t make sense³³, compared to replacing the strings with a hammering low-pitched timpani. One does not need to disrupt timing to disrupt coherence.

The fictional (but realistic) example of a dancer with angry-evoking movements accompanied by a highly congruent happy-evoking music (Fig.1) shows a discrepancy between the syntax-predicted coherence and the incoherence it actually generates. Each modality separately has interpretable semantics, but these feel unrelated and interfere into uninterpretability when put together. Even disjoining the dance and music objects wouldn’t make them fit coherently into a common external situation.

Semantic comprehensibility can predict and nuance coherence of syntactically congruent sequences

Syntactic congruence and semantic comprehensibility can also align in their predictions of coherence, but the combined semantic analysis provides additional insights on how each modality enhances or complements the other coherently. First, dance and music can enhance each other by redundancy of their meanings (Christensen et al., 2014). To interpret the parallel between the musical crescendo and the increase in movement, in Béjart’s “Boléro” (fn.14), as a build up of energy, intensity, and tension that calls for a resolution, we need a semantic dimension – such as conjunction – that can explain how dance and music amplify each other’s inferences.

Second, syntactic congruence doesn’t imply that dance and music must bring redundant information. Semantics can detect their complementarity, especially through interpretations of discourse referents. For example, the syntactically congruent repetition of specific movements with a specific musical motif from Béjart’s “Boléro” (Fig.5) may be interpreted as disjoint reference (Section 4.3.2). Yet, based on congruence, dance and music would be led to point to the same object in a common denoted scene. By acknowledging that dance and music can contribute to a whole with their own - different – semantics, a combined semantic analysis offers alternative interpretations of interactions between discourse referents.

In sum, intuitive coherence of a sequence of combined dance and music is not fully determined by how syntactically congruent the sequence is. Semantics not only nuances syntactic congruence by capturing the complex relationships dance and music can exhibit, but comprehensibility can also reverse coherence predictions to be more in line with our intuitions.

³³This could alternatively have a humorous effect if the context allows it – an interpretation that is also only possible if dance-music combinations indeed have semantics.

5 Conclusion

With the overarching goal of investigating formation of meaning using evidence from multimodal stimuli, this paper attempted to assess the advantages of a semantic theory of combined dance and music over a syntactic theory to explain the intuitive sense of coherence of dance with music. All in all, I conclude that a purely syntactic approach is not expressive enough to capture the richness and diversity of our experience of choreographies and that also considering a semantic analysis is a productive line of research to understand how dance and music combine in our perception.

The complementary roles of syntax and semantics

A syntactic analysis by congruence is useful to identify structural (especially rhythmic) similarities between dance and music. Some instances of dance-music syntactic congruence do indeed yield an impression of coherence, and some instances of syntactic incongruence yield impressions of incoherence. However, on a broader set of choreographical examples, this pattern doesn't always hold. By assessing how a semantic theory fares when faced with these troubling cases, we observed that semantic comprehensibility has the ability to predict coherence in a way that was more in line with our intuitions.

I support that a dance-music semantic analysis is possible and that it provides information as to what situations the dance-music sequence - as an integrated whole - may be true of, by delineating rule-governed patterns of what can or cannot be inferred. These comprehensibility patterns can be modeled by semantic conjunction and discourse referents analyses, as proposed by Schlenker (2022). Conjunction delineates how modalities select for each other's denotations in a way that reduces global underspecification and ambiguity, and, complementarily, the discourse referents analysis formalizes relations between denoted objects not only within modalities but also across modalities.

According to a semantic account, coherence arises from comprehensible semantic relations between dance and music (in line with Lascarides and Stone, 2009). Conversely, a sense of incoherence would arise when there is no semantically acceptable situation to infer from their combined semantics – we are left with two stories depicted in parallel with little connectable information that could be associated within a coherent story. This last point suggests that more narrative choreographies facilitate evaluation of coherence in cases of incongruence (e.g. “Beach Birds”, Fig.2), and of incoherence in cases of congruence (e.g. “Angry-Happy” example, Fig.1) by making semantic relations more evident. In comparison, some people may react to more conceptual, abstract choreographies where dance and music are relatively independent (e.g. “Second Hand”, 2) with dismay, questioning the purpose of following two unrelated stories simultaneously. Semantics thus accounts for a spectrum of coherence across in/congruence, nuancing the rather dichotomic syntactic analysis.

However, syntax and semantics need not oppose each other. The eventuality of competition between syntax and semantic predictions of coherence calls for more nuance than a strictly semantic account. Hypothetically, coherence judgments may be more yoked towards syntax or towards semantics depending on how readily individuals interpret meaning, or salience of structurally congruent elements compared to semantic information. Whether coherence is due to congruence or comprehensibility may therefore depend on the particular choreography (for example, coherence in Balanchine may be more syntax-based, whereas coherence of Cunningham's “Beach Birds” would be more semantics-based).

Limitations

There are nonetheless a number of limitations to the present approach: (1) An alternative syntactic analysis may overcome the shortcomings of our syntactic congruence account of coherence. (2) The

frontier between syntax and semantics is blurry in itself, and the relevance of these linguistic categories for dance and music remains questionable. (3) Dance and music are asymmetrical: dance without music feels stranger than music without dance, there are natural perceptual biases towards auditory or visual stimuli (Alais et al., 2010; Krumhansl and Schenck, 1997), and most dance is choreographed on pre-existing music, creating directional expectations. (4) Contextual information (e.g. costumes, lighting, socio-cultural context) may influence interpretations in a dynamic way (Hubbard, 2019). (5) Perception of music with dance may be influenced by other, non-syntactic and non-semantic, concurrent cognitive processes such as memory or attention (Watanabe and Shimojo, 2005), as well as hedonic evaluation (Howlin et al., 2017; Monroy et al., 2022). (6) Subjective experience, preference, and knowledge may drive what one perceives and interprets (Fogelsanger and Afanador, 2006; Monroy et al., 2022; Vincs et al., 2010). (7) Last but not least, although the goal was to remain general, the very definition of “dance and music” remains vague. These factors could in fact be driving judgments of coherence and also require a more nuanced definition of the latter.

Further research perspectives

A natural next step would be to outline a formal typology of dance-music combinations based on the sequence’s structural and semantic features. Such a typology should take into account dance-music in/congruence, perceptual saliency of different parameters, iconicity with our environment, and integration of previous events in a choreography.

Being based on introspective judgments of coherence and inferences, which are subjective, this research remains to be explored experimentally to have scientific validity. One must first verify coherence intuitions of dance-music stimuli, through participant ratings of a dance-music video, or by asking which combination feels most coherent out of two dance-music pairs. To differentially test for syntactic and semantic effects on coherence, dance-music stimulus pairs should be mixed and matched to individually manipulate syntactic congruence (by applying a temporal lag for example) or semantic comprehensibility (such as by changing emotional valence of the music), using a typology of dance-music features.

More relevant to testing the existence of a semantics specific to dance-music sequences, one can ask participants to choose which out of two situations (non-choreographical images/videos manipulated to be within the conjunct set or not, or to display discourse referents assignments consistent with the joint sequence or not) is best represented by a given dance-music video extract. This can be done in the other direction, by showing participants a situation and asking them to choose which dance-music pair out of two choices (with one dance or music sequence held constant) best represents it. Significantly consistent responses across participants would suggest that dance with music can represent a choreography-external reality in a patterned way. This should be complemented by similarly testing semantic of unimodal sequences in order to determine their contribution to the joint percept. Going further, using brain imaging could shed light on possible neural markers of joint coherence, congruence, and comprehensibility.

Why this matters

The aim of this work is to offer theoretical frameworks of analysis to guide future inquiries of dance and music perception, and more generally of multimodal phenomena, in the fields of linguistics, cognition, and art. For the field of linguistics, comparative studies of communicative processes across modalities beyond language appear to be a productive tool to shed light on elusive syntactic and semantic phenomena, by making certain properties stand out that would not be so clear unimodally. For instance, if dance and music can indeed interact in such a way that they contradict each other (e.g. “Angry-Happy”

example, Fig.1), then it suggests that both dance and music have a semantics. Going further, one could speculate that combinatory semantic theories may be extended as fundamental principles of meaning integration of any combination of perceptual streams, regardless of modality (e.g. music and lyrics in song, or two dancers with different movement patterns).

For cognition, formal accounts can guide experimental research paradigms by better defining structural versus meaning-bearing features, and inform interpretations of often contradictory findings (e.g. the nature of “matching” ratings in Mitchell and Gallaher, 2001). Analyses of multimodal integration using linguistic methodology are productive in understanding the cognitive architecture underpinning human communication, and particularly the overlap in cognitive processing between dance and music at a supramodal level. One may ask where the integration of dance and music that leads to coherence is taking place: (i) pre-linguistically into a supramodal conceptual schema that can then be structurally and semantically processed (Krumhansl and Schenck, 1997; Mitchell and Gallaher, 2001; Saitis et al., 2020; Sammler et al., 2013), (ii) at a syntactic level, (iii) at the level of their resulting semantic inferences (Maffongelli et al., 2015), or (iv) at all three levels. We could then expect that different semantic inferences and coherence intuitions would arise from different integration levels.

So why should someone outside the realm of cognitive science care about this research? A formal framework of multimodal integration can guide analyses of audio-visual interactions in artistic works, as means of communication between an artist and their audience. Aesthetically, the effects observed in this paper provide choreographers with a large range of strategies to create, confirm, or violate expectations, hereby provoking patterns of tension and relaxations by playing with variations in coherence that emerge only when music is put together with dance³⁴.

To conclude, in spite of the controversy surrounding the existence of a semantics beyond natural language, I hope that these explorations have made a case for the necessity of considering multimodal semantics, and substantiated theories of music and of dance semantics, in addition to syntactic approaches. Semantics reveals what dance brings to music and what music brings to dance. Investigations of multimodal integration across levels of analysis are still fresh: there is an imperative for further in-depth interdisciplinary research that builds on formal models, in the hope of better understanding the foundational principles and processes of human means of communication.

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³⁴Informal discussions about this project with composers creating audio-visual works substantiated this perspective.

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Appendix

An analysis of discourse referents in Bausch’s *Ritual of the Rival Tribes* (from “Rite of Spring”)

In Bausch’s “Rite of Spring” (*Ritual of the two rival tribes*, 13:08, fn.12), male and female dancers form two defined groups on stage, each group has characteristic movement sequences associated with two main musical textures and melodic motifs: staccato, low-pitched, and brass-dominated when men dance (13:12-13:44 and 13:51-13:55), and legato and woodwind-dominated when women dance (13:44-13:51 and 13:55-14:09 and 14:44). Dance and music clearly and coherently refer to each other across time even though their non-contiguity renders them independent from a syntactic point of view. These systematic associations determine which group dominates the stage. Such transfers are made possible by semantic assignment of variables (discourse referents) indexing dance-music relations across time so as to refer back to them – such as two rival tribes alternating their actions).

An analysis of musical occlusion in Bausch’s *Augurs of Spring* (from “Rite of Spring”)

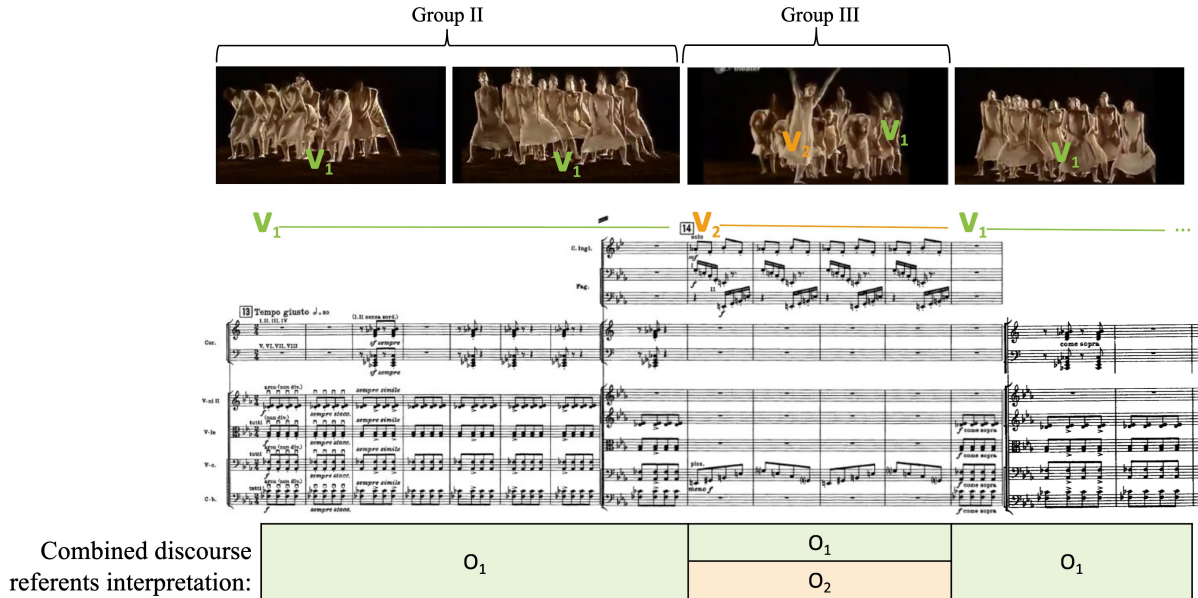


Figure 13: Bausch’s “Rite of Spring”, *Augurs of Spring* (4:29, fn.12), “occlusion” example (introduced in Section 4.3.2): A discourse referents analysis of the dance (top), the music (middle), and a possible semantic interpretation of their combination (bottom) suggests the presence of musical occlusion. Groups II and III are taken from Charnavel’s (2023) syntactic analysis (Fig.3).

In Group II, the discourse referents v_1 from the group of dancers repeating the same movements and the repetitive string ostinato can be interpreted as a multimodal coreference to one object O_1 . At the onset of Group III, the discourse referents v_2 from the solo female dancer who separates herself from the group and the woodwind melodic sequence can be interpreted as a multimodal coreference to another object O_2 . Meanwhile, in Group III, the group of dancers continue repeating the same pattern of movements. Through meaning inheritance, O_1 is still referenced in Group III by the group of dancers, even though the musical ostinato is not playing. As a whole, Group III depicts two denoted objects in the choreography-external situation, with one (O_1) being partially occluded (musically) by the appearance of the other (O_2). At the end of Group III, O_1 fully reappears, as O_2 disappears and the same musical ostinato comes back. This is a striking example, not only of meaning inheritance (stability of a discourse referent across modalities and time) from which a multimodal object is referred to with only one of its modalities, but also of a background-foreground object relationship in a denoted external situation with one partially occluding the other, giving the illusion of continuity of the musical ostinato.